IEC 60204-1 / ISO 13850 EMERGENCY FALLBACK HIERARCHY

Deterministic Fallback Escalation: From Controlled Position Hold to Hard Silicon Safe Torque Off (STO)

CATEGORY 2: CONTROLLED STOP

Normal Operational Pause (Drive Power Maintained)

- Robot decelerates along path to standstill using motor torque
- Mechanical position hold active; power remains on
- Triggered by: Soft geofence warning, task pause, minor latency jitter

CATEGORY 1: DYNAMIC BRAKE

Controlled Deceleration + Power Cut at Standstill

- Motors actively brake at a_max (-4.5 m/s^2) using regenerative/dynamic braking
- Mechanical safety brakes engage once speed $v = 0$, then power is removed
- Triggered by: Expired intent lease, watchdog timeout (50 ms), trajectory divergence

CATEGORY 0: SAFE TORQUE OFF (STO)

Immediate Hardware Power Disconnect (Uncontrolled)

- Dedicated hardware STO relay drops gate drive voltage instantly
- Motors coast to stop under pure mechanical brake / friction
- Triggered by: Physical E-stop button, primary silicon crash, over-current trip